SafeHer

Women Safety Mobile Application

Project Synopsis

Platform: Flutter (Android & iOS) | Category: Personal Safety / Emergency Response
Local-first architecture — no backend server required for MVP

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1. Introduction

Personal safety remains a pressing concern for women commuting, travelling, or living alone. In moments of distress, every second counts, and the ability to quickly alert trusted people with an accurate location can make the difference between a close call and a tragedy. **SafeHer** is a Flutter-based mobile application designed to give users a fast, reliable, and free way to raise an alarm and share their live location with people they trust, without depending on a paid SMS gateway or a hosted backend.

The application follows a local-first design: user sessions, emergency contacts, and profile details are persisted on-device using SharedPreferences, while live location and SOS alerts are handled through the device's own GPS and native SMS/phone capabilities. This keeps the MVP lightweight, free to operate, and easy to extend later with cloud services such as Firebase.

2. Problem Statement

Existing personal-safety solutions often require a data connection to a proprietary backend, a paid subscription, or complex multi-step workflows during an emergency — precisely when a victim has the least time and composure to operate a phone. There is a need for a simple, one-tap safety tool that works using capabilities already built into every smartphone (GPS and SMS), keeping the emergency flow fast and dependency-free.

3. Objectives

- Provide a one-tap SOS mechanism that shares live GPS coordinates with trusted contacts instantly.
- Allow users to maintain a personal list of emergency contacts they can call directly.
- Offer live location tracking that can be viewed or shared via Google Maps at any time.
- Persist login sessions and user data locally so the app works without a constant server dependency.

- Keep the experience minimal, fast, and usable under stress, with clear navigation from a single home screen.

4. Scope of the Project

The current scope covers a fully functional MVP for Android and iOS built with Flutter. It excludes real backend authentication, automatic (silent) SMS dispatch, and in-app map rendering — these are identified as production-hardening next steps rather than MVP requirements. The design intentionally favors the native `sms:` URL scheme over a programmatic SMS API, trading one manual tap for zero infrastructure cost and no third-party SMS gateway dependency.

5. Feature Overview

Feature	Description
Login	Simple phone number and password login screen; session persisted locally so users stay logged in
Home	Central dashboard with a prominent SOS button and quick-access cards to every other feature.
SOS	One-tap emergency alert — captures live GPS location and sends it via SMS to all saved trusted co
Live Tracking	Streams the user's live coordinates in real time and lets them open or share their position on Google
Emergency Contacts	Add, edit, and delete trusted contacts, with the ability to call any of them directly from the app.
Profile	Displays and edits name, phone number, and blood group; also provides the logout action.

6. How the SOS Flow Works

- The user taps the SOS button on the Home screen or the dedicated SOS screen.
- LocationService fetches the current GPS position, requesting runtime permission if needed.
- SosService builds an alert message containing a Google Maps link to that position.
- The device's native SMS app opens, pre-addressed to every saved emergency contact with the message pre-filled — the user only needs to tap Send.

This design keeps the emergency path free of cost and free of a paid SMS gateway or backend server, at the trade-off of one confirmatory tap in the native messaging app.

7. Project Structure

```
lib/
  main.dart                 entry point, routes to Login or Home based on session
  theme.dart                colors & Material theme
  models/
    emergency_contact.dart
  services/
    storage_service.dart    SharedPreferences persistence (contacts, profile, session)
    location_service.dart   geolocator wrapper (permissions + current/live position)
    sos_service.dart        builds SOS message + opens SMS/phone app
  screens/
    login_screen.dart
    home_screen.dart
    sos_screen.dart
    tracking_screen.dart
    contacts_screen.dart
    profile_screen.dart
pubspec.yaml
```

8. Technology Stack

Layer	Technology

Framework	Flutter (Dart) — cross-platform for Android & iOS
Local Storage	SharedPreferences (session, contacts, profile persistence)
Location	geolocator plugin for GPS permissions and position streaming
Emergency Alerts	Native sms: URL scheme + Google Maps link generation
Calling	Native tel: URL scheme for direct calls to trusted contacts

9. Required Native Permissions

Android — declared in AndroidManifest.xml:

- ACCESS_FINE_LOCATION / ACCESS_COARSE_LOCATION — for GPS positioning
- SEND_SMS — to compose the SOS alert message
- CALL_PHONE — to call emergency contacts directly
- INTERNET — for map link generation and future backend connectivity

iOS — declared in Info.plist:

- NSLocationWhenInUseUsageDescription
- NSLocationAlwaysAndWhenInUseUsageDescription

10. Current Limitations

- Login is a local-only placeholder and does not perform real authentication.
- SOS messages require a manual tap to send, since the sms: URL scheme cannot auto-send without native platform code or a paid SMS API.
- Live Tracking currently shows raw coordinates with an 'open in Maps' link rather than an embedded map view.

11. Future Scope

- Replace placeholder login with real authentication (Firebase Auth or a custom backend).
- Add a native platform channel (Android SmsManager) or a Twilio-style SMS API for automatic, tap-free SOS dispatch.
- Integrate google_maps_flutter for an in-app live map view fed by the existing location stream.
- Add a shake-to-SOS gesture for hands-free triggering.
- Introduce a countdown/cancel timer before sending an SOS to prevent accidental alerts.
- Enable background location sharing for the duration of an active emergency.
- Add push notifications and read-receipts so the user knows contacts received the alert.

12. Conclusion

SafeHer demonstrates how a focused set of native smartphone capabilities — GPS, SMS, and calling — can be combined into a fast, dependency-free personal safety tool. By keeping the MVP local-first and backend-free, the project minimizes cost and complexity while still delivering the core promise of the app: getting a user's location into the hands of people who can help, in as few taps as possible. The identified next steps around authentication, automatic SMS dispatch, and in-app mapping provide a clear path from this working prototype to a production-ready release.